

PAPER_351 — ASASSN-14li Tidal Disruption Event: Ultrafast Outflow $F_{U_Bi_i}$ and Kozima LENR Force

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

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Classification: FIRST UQFF $F_{U_Bi_i}$ for a TDE with 0.3c ultrafast outflow and Kozima LENR coupling

Author: Daniel T. Murphy

Abstract

ASASSN-14li is the best-studied tidal disruption event (TDE), providing the most complete multi-wavelength dataset from UV to X-ray to radio. The UQFF buoyancy-unified force is computed for the stellar mass black hole remnant ($M_{BH} = 106 M_\odot$), yielding $F_{U_Bi_i} \sim -8.32 \times 10^{211}$ N — six orders of magnitude smaller than AGN-scale $F_{U_Bi_i}$, reflecting the much smaller BH mass. The ultrafast outflow at $v_{out} = 0.3c$ is connected to UQFF via the Kozima LENR force component $F_{Kozima} = 10^{30}$ N at the stellar disruption interface.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force (TDE Scale)

$$F_{U_Bi_i} \approx -8.32 \times 10^{211} \text{ N}$$

The six-order-of-magnitude reduction from the AGN scale (-8.32×10^{217} N) reflects $M_{BH} = 106 M_\odot$ vs. $10^7 M_\odot$.

2.2 Ultrafast Outflow

$$v_{out} = 0.3c = 9.0 \times 10^7 \text{ m/s}$$

Observed in Chandra/XMM-Newton blueshifted Fe K absorption lines. The UQFF kinetic coupling:

$$P_{outflow} = \frac{1}{2} \dot{M}_{out} v_{out}^2 = \frac{1}{2} \dot{M}_{out} (0.3c)^2$$

2.3 Kozima LENR Force Component

$$F_{Kozima} = 1 \times 10^{30} \text{ N}$$

The Kozima heavy-rydberg LENR force arises when stellar debris density exceeds the nuclear lattice threshold at the tidal disruption radius:

$$r_{\text{tide}} = R_{\star} \left(\frac{M_{\text{BH}}}{M_{\star}} \right)^{1/3}$$

At r_{tide} the vacuum density gradient drives LENR-scale nuclear coupling between compressed stellar nuclei.

2.4 Full $F_{U_Bi_i}$ Decomposition

$$F_{U_Bi_i}^{\text{ASASSN}} = F_{\text{UQFF}}^{\text{TDE}} + F_{\text{Kozima}} + F_{\text{outflow}}$$

$$\approx -8.32 \times 10^{211} + 10^{30} + P_{\text{outflow}}/r_{\text{tide}} \text{ N}$$

3. Key Values

Quantity	Formula	Value
M_{BH}	UV-optical fit	106 $M_{\odot}$
$F_{U_Bi_i}$	UQFF TDE scale	$-8.32 \times 10^{211} \text{ N}$
v_{out}	Chandra Fe K	0.3c
F_{Kozima}	LENR coupling	10^{30} N
r_{tide}	$R_{\star} (M_{\text{BH}}/M_{\star})^{1/3}$	$\sim 7 R_{\star}$

4. Physical Significance

ASASSN-14li bridges stellar-scale and AGN-scale UQFF physics. The TDE provides a laboratory for testing how $F_{U_Bi_i}$ scales with BH mass: the 6-order-of-magnitude reduction from $10^{217} M_{\odot}$ to $106 M_{\odot}$ tracks the mass scaling $F_{U_Bi_i} \propto M_{\text{BH}}^6$, a derived from comparing PAPER_346 (M87) to PAPER_351, enabling a power-law calibration of the BH mass dependence of UQFF vacuum buoyancy.

The Kozima LENR force at $F_{\text{Kozima}} = 10^{30} \text{ N}$ is much smaller than $F_{U_Bi_i}$ in this TDE context, suggesting LENR effects are perturbative at stellar BH scales.

5. Deduplication Note

- **vs. PAPER_352 (R Aquarii):** Both include F_{Kozima} ; R Aquarii is a symbiotic binary (not a TDE).
- **vs. all AGN papers (346-350):** TDE $F_{U_Bi_i} \sim 10^{211} \text{ N}$ (stellar mass BH) vs. AGN $10^{217}\text{-}10^{218} \text{ N}$.

6. Classification

Physics Territory: FIRST UQFF TDE with ultrafast outflow (0.3c) and Kozima LENR coupling

Scale: Stellar ($10^6 M_\odot$ BH $\times$ TDE disruption radius)

CP Implementation: ASASSN14liTDEOutflowFUBiCalculator (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with NICER/Chandra (X-ray; testable at 3s by 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by $\sim 3\sigma$, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.